CHAPTER

15

Direct Filling Gold

Gold foil is undeniably the permanent restorative modality. The term 'gold foil' and 'direct filling gold' is used interchangeably; gold foil is one form of pure gold, other forms are also available.

In teeth with sound pulpal and periodontal health, gold foil is a superior restorative material, especially for small cavities. The proper case selection, adequate cavity preparation and material manipulation may lead direct gold restorations to last for a lifetime.

Direct gold are those gold restorative materials that are manufactured for compaction directly into the prepared cavities.

PROPERTIES OF PURE GOLD

- Pure gold is soft, malleable, ductile and does not oxidize under the normal atmospheric conditions. One grain of gold can be drawn into a wire of nearly 500 feet long. Malleting produces a sheet as thin as 0.13 mm.
- It is yellow in color with a strong metallic lustre.
- Fuses at a temperature of 1063°C and boils at 2200°C.
- Density of pure gold is 19.0–19.3 g/cm^3. (True density is difficult to achieve in restorations because of the voids, which tend to decrease the density of the restoration. The apparent density is in the range of 14–15 g/cm^3.)
- Brinell's hardness number of pure gold is 25. This would seem to contraindicate its use as a restorative material. However, cold working while building the restoration increases its hardness to almost 58–82; known as 'work hardening' or 'strain hardening'.
- Coefficient of thermal expansion is 14.4 × 10^{-6}/°C, which is close to that of tooth structure, i.e. 11.4 × 10^{-6}/°C.
- It has a high thermal conductivity of 0.710 cal/ sec/ cm^2 (°C/cm).
- Gold is the noblest of all metals. Air or water at any temperature does not affect or tarnish gold. It is not soluble in sulfuric acid, nitric acid or hydrochloric acid; however, readily dissolves in *aqua regia* (a combination of nitric acid and hydrochloric acid) to form trichlorides of gold ($AuCl_3$).
- Small amounts of impurities have a pronounced effect on mechanical properties of gold, e.g. 0.2% lead makes gold very brittle. Bismuth and mercury have also shown adverse effects on gold; whereas, addition of small amounts of calcium, palladium and platinum improve its properties.
- Able to be cold worked and welded at room temperature. (If two clean pieces of gold are pressed firmly together and a sufficient force applied, they unite at their point of contact and the gold is welded together without the application of heat.) This process is known as 'cold welding'.
- Provide efficient sealing, when used as a restorative material.

TYPES OF DIRECT FILLING GOLDS

The different forms of direct gold are cohesive and 99.9% pure. It is necessary to understand the properties and indications of each material because the manipulation procedures vary with the choice of material.

The different forms of direct gold are:

I. Gold foil
   i. Gold foil sheets
   ii. Gold foil pellets
   iii. Gold foil cylinders
   iv. Corrugated/carbonized gold foil
   v. Laminated gold foil
   vi. Platinized gold foil
II. Electrolytic precipitated gold (crystalline gold)
   i. Mat/crystal gold
   ii. Mat foil
   iii. Electralloy (gold calcium alloy)
III. Powdered/granulated gold and its combination with gold foil
   i. Goldent
IV. Cohesive and non-cohesive

I. Gold Foil

i. *Gold Foil Sheets*

Gold sheets are manufactured by a process called 'gold beating' or 'rolling'. The light weight sheets are formed by beating and heavy weight sheets by rolling. Heavier mallets are used for beating gold ingots followed by the lighter ones when the gold gets thinner. In the rolling process, gold is passed continuously through the rolling mills until the desired thickness is achieved. The processes mentioned above render gold non-cohesive. To use as a restorative material, gold should be turned cohesive by annealing.

Gold foil sheets are bound and contain 12 sheets; each sheet has the dimensions of 4" × 4" (Fig. 15.1). No. 4 gold foil is the standard size, which is usually used. It weighs 4 grains (0.259 gm/259 mg) and is 0.51 mm thick. Similarly, No.3 gold foil weighs 3 grains, No.2 gold foil 2 grains and so on.

ii. *Gold Foil Pellets*

An ideal pellet is produced from a No. 4 gold foil by the operator. The procedure for fabrication includes removing the No. 4 gold foil, held onto the paper, marked (Fig. 15.2) and cut along with it. The paper supports the thin gold foil and prevents it from tearing and wrinkling during manipulation. Sizes, which are commonly used for rolling into pellets, are 1/16 onwards. Larger sizes like 1/2, 1/4, 1/8 are used for forming cylinders. The desired piece of gold foil is grasped from the center and held between the thumb, index finger and middle finger of the left hand. Ends are tucked and rolled lightly into round balls of gold.

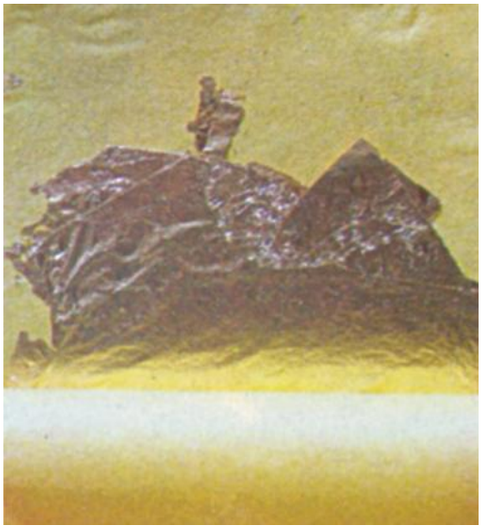


Fig. 15.1: Gold foil between paper leaves

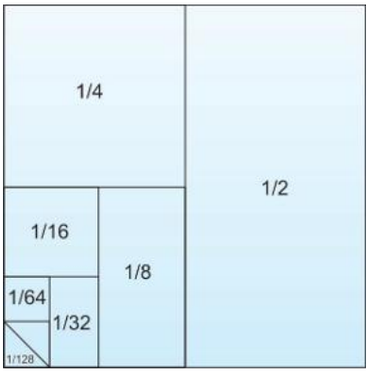


Fig. 15.2: Square showing various sections of gold foil sheet

iii. *Gold Foil Cylinders*

Gold foil cylinders are custom made by rolling cut segments of No. 4 gold foil usually 1/2, 1/4 and 1/8 into desired widths.

iv. *Corrugated/Carbonized Gold Foil*

Corrugated gold foil is manufactured by placing thin sheets of paper in between the gold foil sheets, which are then ignited. The paper burns and gets charred; however, the gold foil remains unharmed, but becomes corrugated because of the shrivelling of paper.

v. *Laminated Gold Foil*

When a gold foil is beaten from an ingot, its crystals are elongated in a specific direction. When two or three foils with crystals running in different directions are combined together, it results in a material that has fibers running in all directions, making it more resistant to the applied forces. Such a combination is known as laminated gold foil.

vi. *Platinized Gold Foil*

This is a sandwich of gold and platinum with the platinum content being 15%; formed in two ways:

- One sheet of pure platinum foil is sandwiched between two sheets of gold foil. These are then hammered until the thickness of one No. 4 gold foil is obtained.
- Layers of platinum and gold are rolled over together so that there occurs fusion of the two, even before the beating procedure has begun.

Platinum increases the hardness and wear resistance of the restoration; used in stress concentration areas.

II. Electrolytic Precipitated Gold (Crystalline Gold)

Precipitated gold consists of the crystalline gold powder formed by electrolytic precipitation. The

powder is formed into different shapes by heating at a temperature well below the melting point of gold.

i. *Mat/Crystal Gold*

The earlier crystal gold is electrolytic precipitated gold obtained by a process similar to electroplating, but at an accelerated rate. During subsequent heating process, branch ends of the crystals are rounded and tend to weld themselves together. These are available in the form of strips of medium widths (2.0 mm) and wide widths (3.0 mm) (Fig. 15.3), which can be cut to the size that fits the cavity. Since mat gold is spongy in nature, it can be easily compacted to the prepared cavity. However, its use is restricted to forming only the internal bulk of the restoration because of its large surface area. Since mat gold is loosely packed, is friable and contains numerous voids, a gold foil is recommended for the external surface of the restoration.

ii. *Mat Foil*

Mat foil is a sandwich of mat gold placed in the sheets of gold foil. The sandwich is sintered by heating below the melting point of gold and cut into strips of different widths. The idea to use mat foil was to eliminate the need to veneer the restoration with a layer of gold foil.

iii. *Electralloy (Gold Calcium Alloy)*

Electralloy is an alloy of electrolytic gold and calcium (0.1–0.5% by weight). The purpose of adding calcium is to produce stronger restorations. The product is sandwiched between two layers of gold foil.



Fig. 15.3: Mat gold of different sizes

III. Powdered/Granulated Gold and its Combination with Gold Foil

Powdered gold is in the form of minute particles, which are obtained by chemical precipitation (gold is dissolved in *aqua regia* and precipitated by oxalic acid, sulfur dioxide or sodium nitrate). The average particle size is 15 μ. The powder is gathered into a conglomerate mass having a diameter of 1.0–3.0 mm. The mass is pre-condensed to facilitate adhesion between the particles and ease of handling. Because powdered gold produces a less porous surface as compared to mat gold, veneering with a gold foil is not necessary.

The pellets of powdered gold wrapped in a gold foil are known as 'Goldent' (Fig. 15.4). The powdered particles held in a gold foil, which acts as a container for the powdered particles and facilitates their condensation. Each pellet contains approximately ten times more gold than a pellet of gold foil of comparable size. The ratio is 95% powder and 5.0% foil.

IV. Cohesive and Non-Cohesive

Pure gold can be classified into cohesive, semicohesive and non-cohesive forms. It is emphasized that two gold surfaces can cohere by welding at room temperature, only if the surfaces are atomically clean. The clean foil can be placed directly into the cavity. Such a form of gold is called the 'cohesive gold'. It is difficult to maintain the cohesive state as some gases may get adsorbed during storage. It is, therefore, recommended to degas even the cohesive forms before placing them in the cavity preparation.

The 'semicohesive gold form' is one in which a protective gas film, like ammonia, is deliberately adsorbed onto the surface of gold by the manufacturer. This is beneficial as it minimizes the adsorption of other less volatile substances.



Fig. 15.4: Goldent (Gold)

This volatile film is removed by heating, making the foil cohesive.

'Non-cohesive form of gold' is one in which certain non-volatile contaminants like iron, sulfur or phosphorus are permanently deposited onto the surface. Non-cohesive forms are occasionally used as a starting material, since they can be worked more easily than the cohesive forms.

The earlier category of semi-cohesive form is now termed as the 'non-cohesive' form; hence, the classification is reduced to two forms: Cohesive and non-cohesive (one in which there is a surface layer of adsorbed ammonia).

ANNEALING/DEGASSING

'Annealing' a gold foil refers to the removal of surface contaminants. (*Degassing' or 'desorption' is the preferred term.)* The primary purpose is to produce an atomically clean surface and render the material cohesive and workable. For non-cohesive gold, degassing removes the protective ammonia film. In cohesive gold, various contaminants like, sulfur and phosphorus compounds, oxides or water vapour may have been adsorbed onto the surface during storage. Temperatures in the range of 600–1300°F is required to volatilize gases or other contaminants.

The commonly used methods for degassing are:

i. ***Alcohol flame:*** This is the preferred method for degassing. Absolute/90% ethyl alcohol is used to produce a clean blue flame. Regular cohesive gold is held only momentarily for a few seconds in the flame until the color changes to dull cherry red. Powdered gold must be held in the flame until all the wax is burnt out and the pellet reaches a dull red glow.

Degassing may be carried out by piece method or bulk method. In the piece method, a single piece is annealed at one time. The gold is held with a suitable instrument, usually a nichrome wire (Fig. 15.5A).

Advantages

- Minimum wastage
- Ability to select a piece of desired size
- Eliminates contamination amongst different pellets during their heating and use

'Bulk degassing' is the degassing of several gold pellets at the same time. Mica trays are used to hold the gold over the flame. Bulk annealing has the advantage of taking less time in the whole process compared to the piece method. However, gold pellets may stick together, if the tray is accidentally moved during heating.

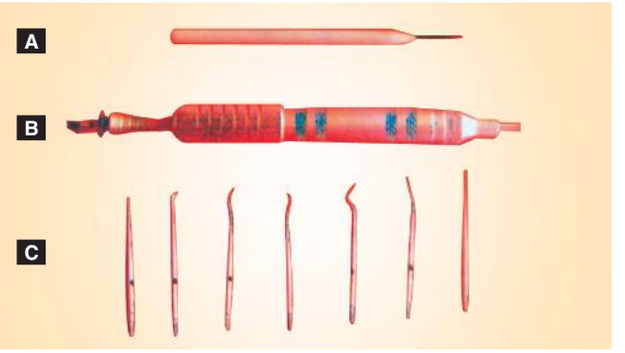


Fig. 15.5: Instruments used in a direct filling gold restoration: **(A)** Nichrome wire used for carrying gold; **(B)** Pneumatic condenser; **(C)** Detachable condensing tips

Precautions taken during flame degassing are:

- The lamp should not run out of fuel during the procedure.
- There should be no surface contaminants and waxes on the lamp and wick.
- Wick should be properly trimmed and rounded to produce a tear drop flame.
- Exercise care during lighting as sulfur in the matches could adhere to the wick and contaminate the gold (ignite with the other end of the match stick).

ii. ***Electric annealer:*** The desired amount of gold is placed in the tray of the annealer and the lid closed. The gold is heated at a temperature of 800°F for 10 minutes and then allowed to cool before it is placed in the cavity.

Disadvantages of electric annealer:

- Inadequate/excessive amounts may get degassed.
- Any remaining gold needs to be discarded because of possible irreversible contamination.
- A temperature of 800°F may not be adequate (require a temperature of 900°F–1200°F).

Care should be taken to avoid underheating or overheating during degassing. 'Underheating' does not adequately remove all surface impurities. This results in incomplete cohesion, because of remaining contaminants leading to overall inferior properties of the restoration. 'Overheating' leads to excessive sintering and possible contamination from the tray, instruments or flame.

CONDENSATION/COMPACTION

Condensation is the procedure used to condense and harden gold inside the cavity preparation. Proper condensation results in a dense, non-porous gold restoration. Each pellet must be thoroughly condensed before the next is added and condensed, otherwise may result in porosity.

The objectives of compaction are:

- Cohere two pieces of gold to each other. This process is known as welding.
- Adapt the gold intimately to the walls and margins of the prepared cavity.
- When the gold is condensed, slip planes are developed and the resulting stresses harden the restoration.
- To drive away air from in between the pieces of gold and gold–tooth interface.

The compaction techniques vary with the types of gold. The features affecting compaction are:

i. ***Force of the blow:*** The average force required to condense gold with a condenser point of 1.0 mm diameter is 15 pounds. It has been found that this much amount of force can be obtained only when the operator applies maximum finger thrust. This would imply that the operator gets rapidly tired, if hand pressure alone is used for condensation. The requisite force from a mallet is recommended because it is rapid, easier and less strenuous. Blows from the mallet should neither be too strong nor too light. Stronger blows deliver heavier forces which could perforate the gold and damage pulp and supporting structures. With lighter blows, inadequate force is applied, which could result in imperfect compaction, poor adaptation to cavity walls and a restoration full of voids. Factors which control the ultimate force delivered in a blow are:
   - Weight of the hammer and the velocity with which it is brought down on the condenser
   - Resistance of the substance receiving the blow
   - Area of the condenser point in contact with the restoration.

   For the first two characteristics, force is directly proportional to the weight and velocity of the mallet.

   Size of the condenser nib, i.e. diameter and area also influence the amount of total force needed. Larger diameter condenser nibs require greater forces for compaction as compared to smaller nibs. It is established that when the diameter of the instrument is doubled, forces required are increased four times.

ii. ***Angulation of condenser (line of force):*** Force applied with a condenser parallels the long axis of the instrument shaft regardless of the angle of the working point. While compacting in the center of the cavity, forces are usually directed at 90° to the pulpal floor. Black advocated an angle of 45° to the cavity walls at the periphery, such that the line of force bisects the line angles and trisects the point angles.

iii. ***Size of condenser tips:*** Small condenser tips (0.5–1.0 mm) are usually considered adequate to achieve the desired compaction. Very small diameter tips tend to perforate the gold. Large diameter tips require high forces for compaction and are less effective in forcing gold into the line and point angles.

iv. ***Method of stepping the instruments:*** Uniform stepping of the condenser point is important to ensure hardening and welding of the entire restoration without leaving any voids. *Stepping is the overlapping of the previous area of the condenser's stroke by half or one-fourth both in individual steps and in lines of steps* (Fig. 15.6).

   Stepping process can proceed in two ways:
   - Moving parallel to the wall and wedging the final row between the already condensed mass of gold and cavity wall.
   - Moving perpendicular to the wall. The final row wedges the gold between that wall and the already condensed gold.

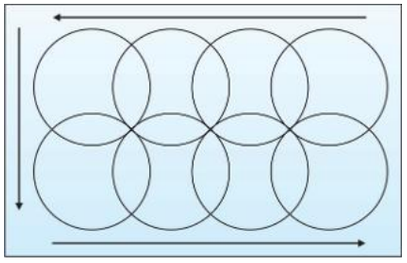


Fig. 15.6: 'Stepping' pattern of the condenser

Condensers

Condensers/pluggers are instruments used to deliver the forces of compaction to the underlying restorative material. There are several methods for the application of these forces:

a. ***Hand pressure:*** Hand pressure is contraindicated except in a few situations, like adapting the first piece of gold to the point angle. Powdered gold can be better condensed with hand pressure. Small condenser tips of 0.5 mm in diameter are generally recommended (create adequate force).

b. ***Hand malleting:*** Condensation by hand malleting is a team work in which the operator directs the condenser and moves it over the surface, while the assistant provides rhythmic blows from the mallet. Long handled condensers and leather faced mallets are used for this purpose. The technique allows greater control; however, with the introduction of mechanical malleting, use of this method has decreased considerably.

c. ***Automatic hand malleting:*** This method utilizes a spring-loaded instrument that delivers the desired force once the spiral spring is released. The disadvantage is that the blow descends rapidly,

even before the pressure is exerted on the condenser point.

d. *Electric malleting (McShirley electromallet):* This instrument has a mallet in the handle, which remains dormant, until the operator put it to function. The intensity/amplitude generated can vary from 0.2 ounces to 15 pounds and the frequency can range from 360–3600 cycles/ minute.

e. *Pneumatic malleting (Hollenback condenser):* Pneumatic mallets consist of vibrating condensers and detachable tips (Fig. 15.5B and C) run by compressed air. The air is carried through a thin rubber tubing attached to the handpiece. The air pressure is controlled by a rheostat, which allows adjusting the frequency and amplitude of condensation strokes. Pneumatic mallets are available with both straight and angled handpieces.

To obtain proper condensation, causing least discomfort to the patient, a small amount of hand pressure is combined with malleting. The condenser point is placed firmly in contact with the gold using slight pressure. It is only then that the blows from the mallet are applied. Teeth accustomed to heavy masticatory stresses would be able to bear heavy malleting forces. Multi-rooted teeth withstand greater condensation pressure than single-rooted teeth. Restorations placed on areas of heavy occlusal stresses, such as incisal edges and cusp tips, require heavy condensation. It is suggested that regardless of the type of condensation method used, the desired compaction should be achieved with forces that do not damage the tooth structure.

Hand condensers have a long handle (six inch) and a blunt end; whereas, mechanical mallets have short shanks of one inch in length. Condenser shanks may be straight, monoangle or offset. Nibs of the pluggers can be of various shapes like round, rectangular and parallelogram. The latter two condensers compared to the former, require three times more force during compaction because of their large size.

The wedge-like serrations present on the face of the condenser nibs increase their surface area. The serrations prevent slipping of the gold and to cut through the outer layers to allow entrapped air to escape. When using angled pluggers for condensation, the effective force is decreased, because the force does not act at right angles to the tip surface. The lateral force may displace the placement of tip or displace the gold.

Compaction/Condensation Technique

The gold foil cut to the size and shape of the cavity dimension is delicately spread out by hand. Gold is adapted lightly to the cavity walls, line angles and point angles by hand pressure. A condenser of desirable size is used to begin malleting in the center of the mass. Slowly, the condenser is moved to the periphery along straight lines using the stepping process. As it reaches the walls, the condenser is turned from 90° to 45°, such that it bisects the line angles and trisects the point angles (Fig. 15.7A, B). While building the restoration, forces of condensation must be directed at 90° to the previously condensed gold; prevent shear components from displacing the already condensed pieces of gold.

The compaction of 1st layer produces a building shelf and thereafter a gold bank is developed onto it. The gold bank is kept against the walls, until the cavosurface margins are reached, thus creating a concave surface while building the restoration (Fig. 15.8). The central concave area left in the restoration is filled last. In case the gold buldges in the center, it may prevent proper application of the force and hence, poor adaptation. Improper condensation could lead to 'bridging' of gold. Bridging is the covering up of small crevices and pits present in the deeper portions of the cavity, especially adjacent to the cavity walls.

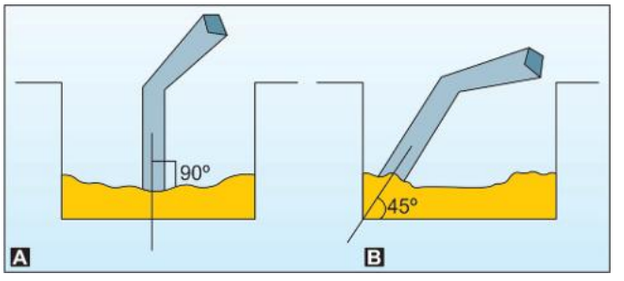


Fig. 15.7: Compaction/condensation of direct gold: **(A)** Condenser held at 90° to the pulpal floor in the center of the cavity preparation; **(B)** Condenser held at 45° when it is moved towards the periphery of the cavity preparation, such that it bisects the line angle and trisects the point angle

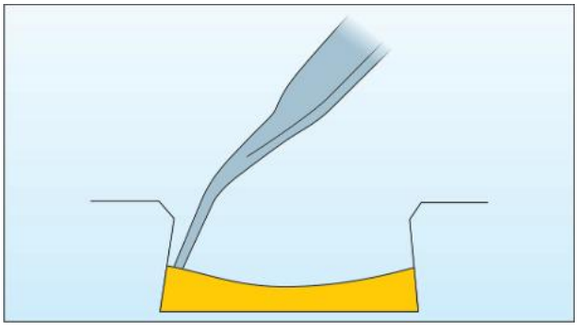


Fig. 15.8: Gold banked against the walls ahead of the center, thus creating a concave surface

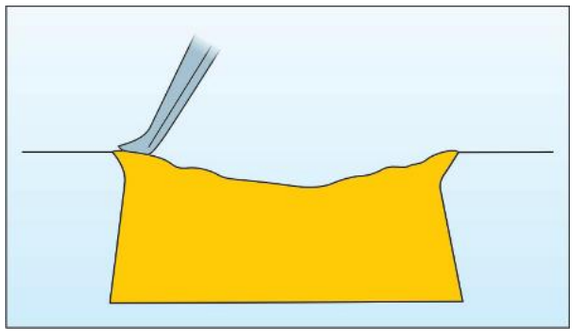


Fig. 15.9: Condensing gold in the cavosurface area

Crushing the cavosurface enamel is a frequent mishap. To avoid this, some amount of gold should invariably be present between the condenser point and the enamel margins (Fig. 15.9). If such an injury is recognized early in the process, removal of the sheared components could re-establish a clean margin allowing condensation to be completed. Retention in a direct gold filling is best obtained by harnessing the elasticity of dentin. During condensation, forces tend to compress the dentin, so that when the filling is completed the dentin returns back to its original position, thereby tightly holding the restoration.

Condensation of Gold Foil

Because of the toughness and smaller mass per unit volume, gold foil can be easily manipulated in smaller cavity preparations and against cavity bevels. The property of the material to roll out of the undercuts towards the condenser may pose problem. This requires making sharp line and point angles to serve as retentive locks and using holding instrument to secure the gold in position while it is being condensed. If pellets are used, the first gold pellet is condensed into a point angle (Fig. 15.10). A holding instrument is used to stabilize the previously condensed gold, while additional pellets are added.

Condensation of Mat Gold

Mat gold is cut to the shape and size of the cavity and placed on the pulpal/axial walls of the preparation. It is condensed against the walls using two condensers. One condenser stabilizes the gold, while the other performs condensation. Two layers of gold are of sufficient thickness to resist curling of the gold during subsequent condensation. Best results are obtained when hand pressure condensation is followed by malleting. Mat gold is easy to start because of their loose spongy nature (known as 'starter gold'). The thickness of mat gold may create problems. If the gold is too thick, it may get strain hardened earlier, preventing the condensation forces affecting the deeper portions.

Condensation of Powdered Gold

The cavity preparation for powdered gold differs slightly; (i) It does not require sharp line angles and point angles, (ii) bevels need to be minimized, because powdered gold condenses more readily against butt cavosurface margins. Upon contact with a condenser, the pellet of powdered gold tends to break apart and wedge itself into the opposing undercuts. It does not require a holding instrument for stabilization. Powdered gold should be firmly wedged between two opposing walls for proper condensation. A pellet of larger size or two/three pellets may be placed together to wedge the space. Pressure is then exerted with a rocking motion to adapt and harden gold initially in the line angles (Fig. 15.11). After the periphery has been built by powdered gold, the center of the preparation is then filled.

BIOLOGICAL EFFECTS OF GOLD FOIL RESTORATION

The pulpal health following a direct gold restoration is questionable because of high forces of malleting. It has been established that the intensity of pulpal reactions increase significantly after direct gold restorations; however, resolution occurs within 4–5

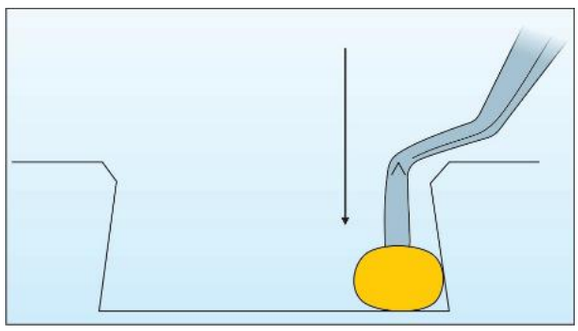


Fig. 15.10: First gold pellet being condensed into the point angle

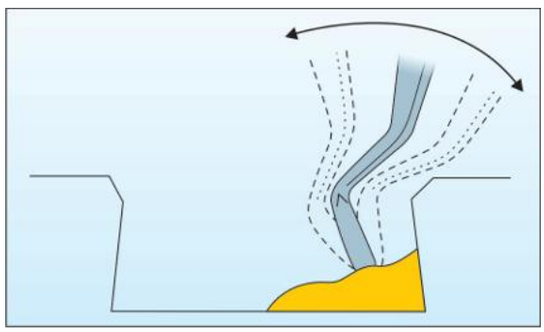


Fig. 15.11: Rocking motion of the instrument to condense the gold

weeks by the regeneration of odontoblastic layer and deposition of cementum. (Pulpal reaction includes displacement of odontoblasts, areas of internal hemorrhage and vesicle formation.)

It is established that with an effective remaining dentin thickness of 3.0–3.5 mm, a normal pulp will undergo a healthy reparative action.

An effective remaining dentin thickness of less than 1.0 mm may lead to destruction of pulp dentin organ. Hence, a direct gold restoration is contraindicated in these situations.

Indications for Direct Gold Restoration

i. ***Incipient carious lesions:*** The direct gold should be limited to only small cavities. Because of its property to get easily deformed under forces, pure gold does not bear the masticatory stresses adequately. The indications include:
- Class I lesions in premolars (lingual pits on maxillary anteriors and facial pits on molars)
- Small Class II lesions (premolars)
- Class III lesions (maxillary and mandibular)
- Class V lesions (small lesions on the facial surfaces of premolars and canines). Not advised in maxillary incisors and molars, because of limitations of esthetics in the former and poor accessibility in the latter.

ii. ***Erosion lesions:*** Same as those mentioned for Class V lesions.

iii. ***Atypical lesions***
- Proximal lesions on teeth adjacent to crown preparations.
- Vent holes in crowns and defective inlay (crown margins can be effectively repaired with direct gold).
- Class VI lesions (cusps tips and incisal edges) (Fig. 15.12A, B).
- Retrograde root canal filling.

Contraindications for Direct Gold Restoration

- Teeth with very large pulp chambers (susceptibility to get damaged during condensation).
- Periodontally weakened teeth (unable to withstand the condensation forces).
- Large carious lesions (remaining dentin may be inadequate).
- Handicapped, elderly or young patients (unable to withstand long sittings).
- Psychologically weak patients who fear the malleting process.
- Inaccessible/poorly accessible areas.
- Isolation difficulties.
- Areas of undesirable occlusal stresses.
- Patients with a high caries index.
- Hypoplastic areas, as forces required to condense gold against the enamel margins might fracture the latter.
- Financial constraints.

Advantages of Direct Gold Restoration

- A permanent method for repairing teeth, and the restoration can last as long as the tooth itself.
- A noble metal, hence does not tarnish and corrode in the oral cavity.
- Insoluble
- Coefficient of thermal expansion is close to that of tooth.
- Small cavity preparation is atraumatic to dental pulp and the supporting structures.
- Tooth discoloration does not occur around the margins because of good material adaptation and its noble and inert qualities.
- No cementing medium is necessary for retention of the restoration.
- The surface can be effectively polished and the smoothness can last indefinitely reducing the chances of plaque accumulation.

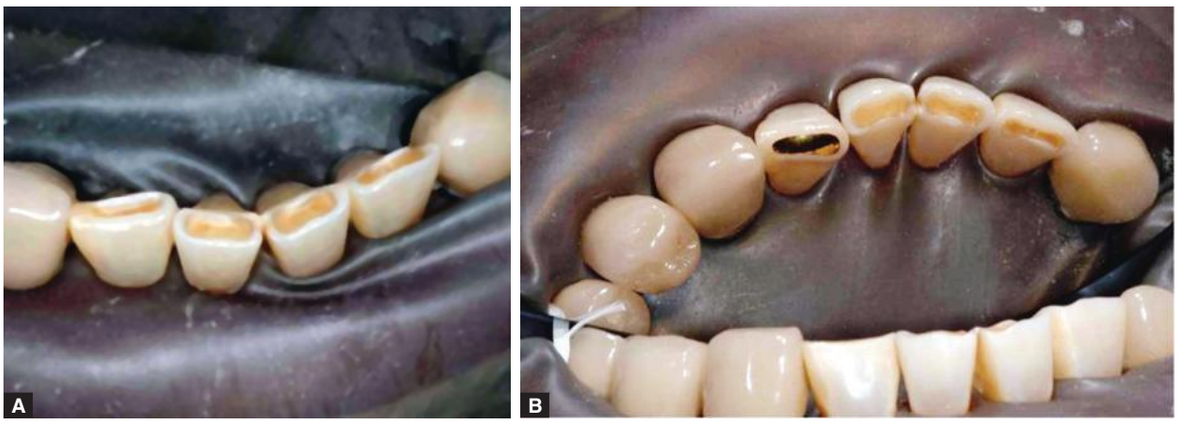


Fig. 15.12: Gold foil restoration in Class VI cavities: **(A)** Attrition/incisal caries in mandibular incisor teeth; **(B)** Restored mandibular lateral incisor with gold foil

- Pure gold is ductile and can be burnished to produce an accurate margin for the restoration.
- Density and hardness of compacted gold adequately enables the restoration to withstand compressive forces.
- Restoration procedure may develop skills of the operator in other dental procedures also.
- Most efficiently sealing permanent restorative material.

Disadvantages of Direct Gold Restoration

- Long chairside time required for restoration
- Use limited to conservative cavities
- High thermal conductivity could lead to post-operative sensitivity
- Technique sensitive
- High condensation forces may injure the tooth and supporting structures
- Expensive
- Unesthetic

CAVITY PREPARATION AND RESTORATION

It is mandatory to apply a rubber dam for proper isolation during gold foil restoration. It should be noted that even the breath of the patient could contaminate the surface of gold and render it non-cohesive.

Class I Cavity

The cavity is kept as small as possible consistent with suitable access for instrumentation and manipulation. In case of a pit defect, the outline may be simply triangular, oblong or circular (Fig. 15.13).

The basic form of cavity simulates that for silver amalgam (Fig. 15.14). Internal outline form includes extending the pulpal floor to 0.5 mm below the dentinoenamel junction.

Retention is provided by the parallelism of walls and sharp line angles and point angles. The angles can be accentuated using No. 6½–2½–9 hoe. Additional retention, if desired, is given in the form of undercuts in dentin by suitable inverted cone bur

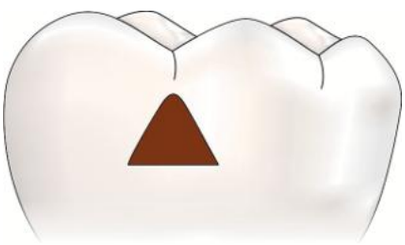


Fig. 15.13: Cavity preparation for direct gold restoration (buccal pit)



Fig. 15.14: Class I cavity prepared on mandibular premolar

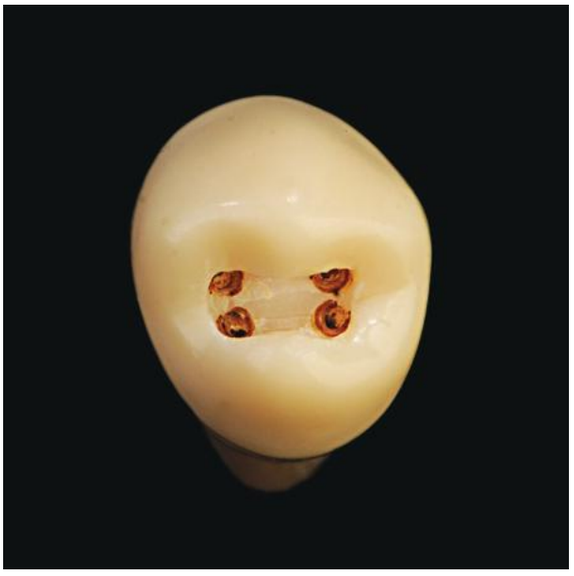


Fig. 15.15: Retention grooves placed facially and lingually in a Class I cavity preparation for direct filling gold

or a No. 6½–2½–9 hoe. These retention grooves are placed facially and lingually in the occlusal preparation of posterior teeth at the faciopulpal and linguopulpal line angles, respectively (Fig. 15.15). For the lingual surfaces of anterior teeth and facial surfaces of molars, these are given incisally and gingivally on inciso-axial and gingivo-axial line angles, respectively.

A very slight cavosurface bevel of 30–40° and 0.2 mm in width is placed on all the margins using a smooth finishing bur. Bevels aid in removing any rough enamel and create thin metals for ease of finishing.

This is the standard cavity preparation for restoration with gold foil or mat gold. Variations in design for powdered gold include roundeninng the line and point angles and establishing butt joint cavosurface margins.

Restoration

The cavity is usually filled with mat gold and veneered with gold foil; may be entirely filled with gold foil or powdered gold also.

i. ***Restoration utilizing mat gold veneered with gold foil:*** Cavity varnish is applied on all the preparation surfaces. A piece of mat gold is cut to a size slightly larger than the cavity preparation, degassed and placed into position by a carrying instrument. Using two condensers, gold is spread out in the form of a blanket and intimately adapted to the preparation surfaces. Malleting is not advised at this stage as the purpose of this step is to simply adapt the gold. It is only after the gold has been properly adapted, malleting is begun in the center at 90° to the pulpal floor. The condenser is moved by stepping process towards the line angles to complete the compaction. On reaching the periphery, the line of force is changed to 45° to the cavity walls. Another increment is then added and condensed in a similar manner until the cavity is half to two-thirds full.

The remaining cavity is completed with gold foils. The gold foil pellet (size 1/16, 1/32, 1/64) is degassed and placed into the cavity along the wall-restoration interface. It is first spread and secured against the already condensed mat gold with hand pressure and then mallet compacted. Cavosurface margins are covered with gold as soon as possible. Gold should always be present between the condenser face and marginal enamel, otherwise a direct blow from the plugger might injure the enamel. The central concave area is finally filled with gold to a slightly higher level. An excess on the surface allows for finishing and polishing procedures.

Burnishing is performed with a special spratley burnisher/a flat beaver tail burnisher/ball burnisher that is run with considerable pressure from the restoration to the tooth surface (Fig. 15.16). Contouring is done with a green stone/garnet coated discs (Fig. 15.17). Gross excess at the margins is removed with a cleoid discoid carver or a gold foil carver (Fig. 15.18). After the excess has been contoured and removed, burnishing is again performed. A high lustre is imparted to the restoration by using powdered abrasives, like tin oxide, pumice, etc.

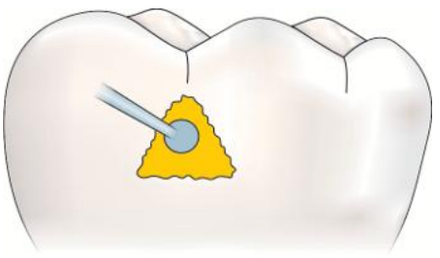


Fig. 15.16: Ball burnisher is used to rub the entire surface

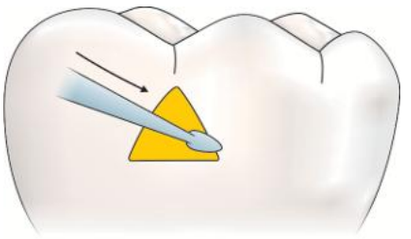


Fig. 15.17: Gold foile is used with push-pull motion to develop the contour

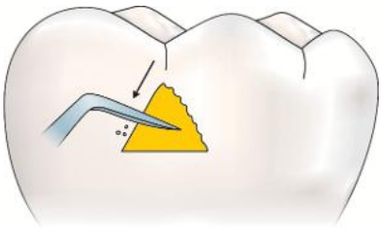


Fig. 15.18: Marginal excess or tags are clipped off with gold foil carver

ii. ***Restoration utilizing only gold foil:*** When gold foil pellets are used to completely fill the cavity, the first pellet is placed in the most distal point angle (e.g. disto-bucco-pulpal point angle). Using a holding instrument, the gold is firmly wedged with hand pressure between the three surrounding walls. Malleting forces are delivered to compact the gold. The second pellet is added onto the first one along the distopulpal line angle. This pellet is also first adapted by hand pressure followed by malleting in the disto-bucco-pulpal direction. Additional pellets are similarly condensed until two-thirds or more of the line angle is covered.

The rest of the pellet is wedged into the distolinguopulpal point angle using forces in a distolinguopulpal direction. This creates a bar of gold running between two point angles.

The concavity in the center is filled with medium- to large-sized pellets by directing forces in the four corners separately. Additional increments are similarly added in a concave form and the cavosurface margins covered with gold. The remaining depression in the central area is finally restored to a slight excess. The completed restoration is burnished, contoured and polished (Fig. 15.19A to G).

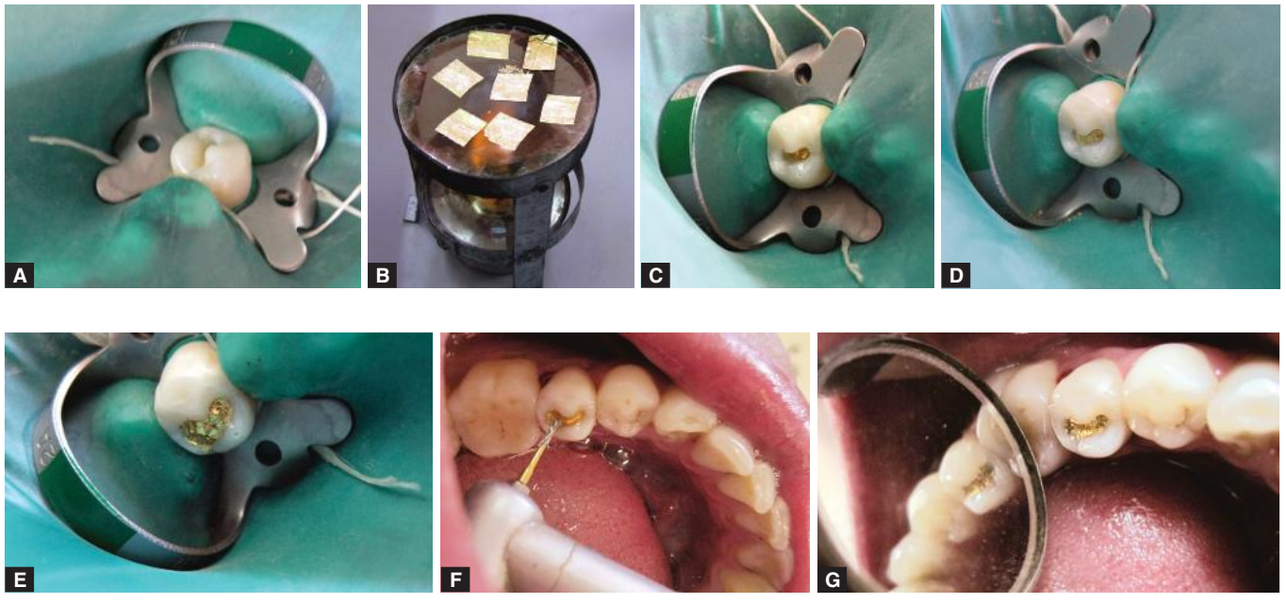


Fig. 15.19: **(A)** Class I cavity prepared on mandibular premolar; **(B)** Heat treatment of gold pellets; **(C and D)** Condensation of gold foils; **(E)** Condensation completed; **(F)** Burnishing and finishing; **(G)** Completed Class I restoration

Class II Cavity

The outline form is similar in size and shape to the preparation for amalgam. The occlusal portion of the Class II preparation is similar to the Class I cavity.

The proximal box is placed in dentin and is in the form of a cone. The width of the proximal box at the gingival portion is only slightly greater than at the occlusal portion. All the walls of the proximal box are extended minimally in their respective embrasures.

Retention is provided by sharp line angles and point angles (differs from an amalgam preparation, which has slightly rounded angles). These angles are accentuated with Hoe and gingival marginal trimmers. Cavosurface bevels of 30–40° are placed on all the margins (compared to an amalgam preparation, which ends in butt joints) (Fig. 15.20A).

Restoration

A matrix is necessary for restoring Class II preparations which would serve as a proximal support. A small shim steel matrix band, which extends 1.0 mm above the marginal ridge and 2.0–3.0 mm beyond the proximal contact is placed interproximally. The matrix is secured with a wedge. Acrylic resin in the doughy stage is adapted and wedged into the facial and lingual embrasures, such that the matrix does not move.

Restoration of the cavity is carried out using a combination of mat gold and gold foil.



Fig. 15.20: **(A)** Preparation of Class II cavity; **(B)** Restored Class II cavity

After the cavity has been completely filled, the acrylic is cut through and the two portions pushed apart. Wedge is removed followed by the band. The restoration is burnished, contoured and polished (Fig. 15.20B).

Class III Cavity

For preparing Class III cavity, numerous designs, such as Ferrier design, Loma Linda design, Ingraham design, Lund, Baum and Woodbury design have been suggested. Prior to any instrumentation or restoration of the cavity, separation is carried out with a Ferrier's separator. A minimal separation of 0.25–0.50 mm is considered adequate.

The routinely followed designs are as follows.

i. *Ferrier Design* (Fig. 15.21)

This design is indicated for small carious lesions with sufficient amount of labial, lingual and incisal walls. Ferrier's design is commonly used on the distal surfaces of anterior teeth, which are less conspicuous as compared to mesial.

The general outline form is roughly triangular in shape. With a No. 1 round bur, the carious lesion is approached from the facial surface. A suitable inverted cone bur is used to extend the marginal outline form.

From the facial view, gingival 4/5th of the facial margin is straight and parallel to the external contour, and in the incisal 1/5th, it curves gently to meet the incisal margin. When viewed proximofacially, the facial outline follows the buccal contour of the tooth and meet the gingival margin in a slight obtuse angle. Gingival margin is straight faciolingually and placed beneath the free gingiva, if possible.

The incisal margin is extended incisally to the contact area, providing access for instrumentation.

The axial wall extends 0.5 mm into dentin and is flat faciolingually and incisogingivally. Retention is provided by the inwards sloping dentinal portion of the gingival wall, i.e. an acute axiogingival line angle and an incisal undercut placed into dentin directed facioincisally. A small bibeveled hatchet is used for establishing the incisal undercut. Small angle formers are used to sharpen the linguoaxiogingival and facioaxiogingival line angles and for creating the acute axiogingival line angle.

A bevel is placed on all the enamel margins using a Wedelstaedt chisel.

ii. *Loma Linda Design*

The Loma Linda design favors a lingual access. It is indicated in those areas where esthetics is the major concern.

Cavosurface bevels and flares are not needed for these preparations and the walls should terminate in butt cavosurface margins. Retention is provided by grooves in three opposing directions using a suitable round bur. A suitable inverted cone bur may be used for placing incisal retention area in the incisal point angle. The other two grooves in the form of a cylinder (1.0–1.5 mm deep) are given on the gingival wall.

iii. *Ingraham Design*

Ingraham design is indicated primarily for the incipient proximal lesions in anterior teeth where esthetics is the major concern. Maintenance of oral hygiene and low caries and plaque indices are essential requirements when using this design.

The general form of the cavity is parallelogram in shape.

Retention grooves are placed on the incisoaxial and gingivoaxial line angles, such that their depth decreases towards the lingual surface. This would

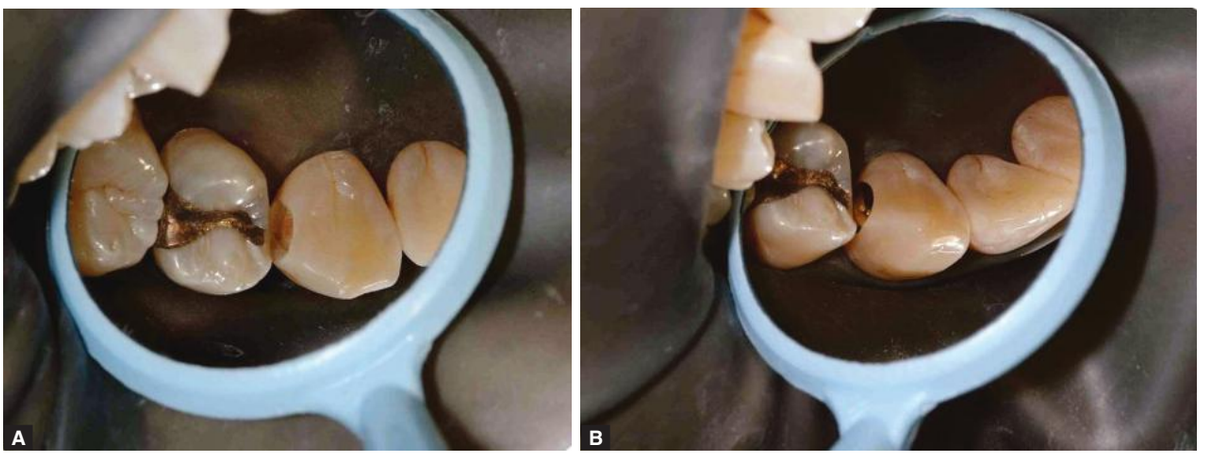


Fig. 15.21: Class III (Ferrier design): **(A)** Preparation of Class III cavity; **(B)** Restored Class III cavity

mean that the undercuts are deeper towards the labial position of the line angle as compared to the lingual portions.

Bevel may be placed on all the margins using appropriate hand instruments.

Restoration

The prepared cavity is filled with mat gold along with veneering gold foil as described.

After the cavity is filled two-thirds, the incisal portion is restored. Building the incisal area is known as 'making the turn' and is accomplished as follows: (i) Sufficient gold is compacted on the lingual wall, such that it is near to the incisal angle, (ii) Small-sized pellets (1/128) are used to fill the incisal area with hand condenser, (iii) Gold foil pellets are compacted into the linguoincisal and incisal area with an offset condenser.

Polishing is avoided as a surface devoid of high lustre will reflect less light and be more esthetic (Fig. 15.21).

Class V Cavity

The most commonly followed Class V cavity design is the Ferrier's design. External outline is trapezoidal in form. Straight lines are preferred in the preparation as these are readily discernible when removing excess gold at the margins.

Occlusally, the margin is straight and parallel to the occlusal plane of the tooth. It extends beyond the carious lesion to sound tooth structure and is limited mesiodistally by the proximal line angles. Similarly, the gingival margin is straight and parallel to the occlusal plane of the tooth but shorter than the occlusal margin. It extends gingivally beyond the carious lesion and mesiodistally limited to the line angles of the tooth.

The axial wall is straight occlusogingivally and parallel to the facial surface of the tooth. Mesiodistally, it has a slight curvature similar to the external contour of the tooth. Axial wall forms a sharp right angle with the occlusal wall, an acute angle with the gingival wall and obtuse angles with the mesial and distal walls. The divergence of mesial and distal walls towards the surface prevents undermining the enamel.

The acute axiogingival line angle is the key to retentive feature. It is prepared with a No. 6½–2½–9 hoe, working from the cavosurface to the axial wall. Sharp line angles and point angles aid in retention. Starting points or convenience points in the form of coves or triangles are prepared incisally and gingivally at the point angles using suitable inverted cone bur. A slight cavosurface bevel is given on all the margins using a Wedelstaedt chisel. A bevel on the gingival wall is avoided, especially when gingival wall extends to the cementum.

Variations in Design

a. When esthetics is the major demand, e.g. in anterior teeth, the occlusal outline may be curved to follow the gingival contour (Fig. 15.22A).

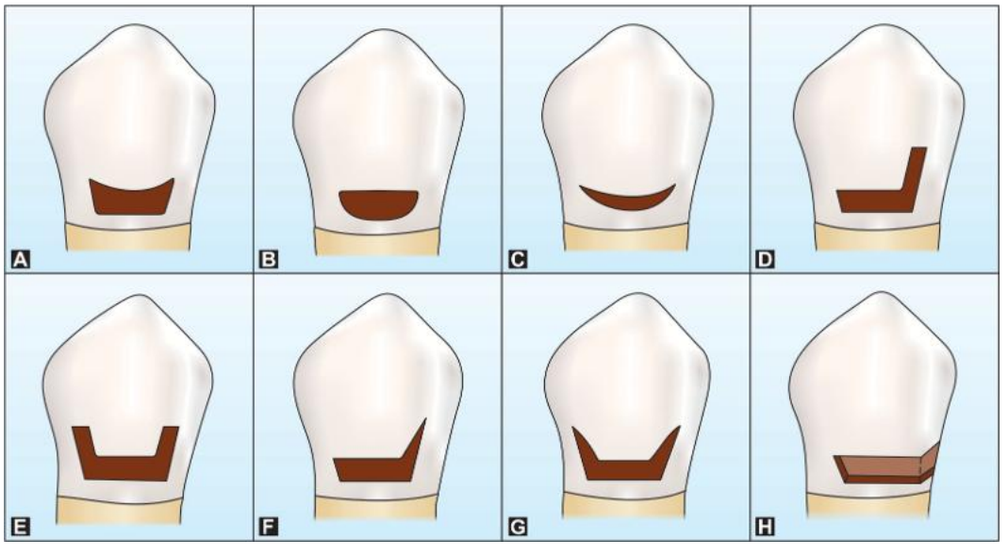


Fig. 15.22: Variations in Class V cavity design for direct filling gold: **(A)** Curved occlusal outline to display less gold; **(B)** Curved gingival outline when gingival tissue is strong and adherent; **(C)** Partial moon-shaped cavity preparation; **(D)** Basic trapezoidal form with a unilateral step; **(E)** Basic trapezoidal form with bilateral steps; **(F)** Basic trapezoidal form with a unilateral moustache extension; **(G)** Basic trapezoidal form with bilateral moustache extensions; **(H)** Class V cavity with parallelogram extension on proximal surface



Fig. 15.23: **(A)** Preparation of Class V cavity; **(B)** Condensation of gold foil; **(C)** Restored Class V cavity

b. Thin strongly adherent gingival tissue may make it necessary to prepare a curved gingival wall (Fig. 15.22B).
c. At times because the height of contour is located apically, the gingival margin is curved in continuation with mesial and distal margins, as if creating a circle. The occlusal outline is also made to follow the contour of the gingiva, so the general shape looks crescent shaped (Fig. 15.22C). This form of cavity is commonly indicated for maxillary and mandibular canines and upper first premolars.
d. Sometimes caries or decalcification extends from the gingival lesion towards the occlusal surface on the proximal line angles of the tooth. In such cases, extension(s) may be added to the basic trapezoidal form in the form of a step (Fig. 15.22D to G), rather than carrying the whole of the occlusal wall occlusally.
e. Occasionally, a carious lesion may extend from the facial surface to the proximal surface(s). In such situations, the basic Class V cavity is prepared on the facial surface with a parallelogram-like extension on the proximal surface (Fig. 15.22H).

Restoration

Cavity varnish is applied thoroughly on all the preparation surfaces. Pieces of mat gold/mat foil/ electralloy is used for forming the bulk followed by veneering with a gold foil. The cavity may be filled completely either with gold foil or powdered gold (same as for Class I cavity) (Fig. 15.23A to C).

BIBLIOGRAPHY

1. Alperstein KS, Yearwood L, Boston D. E-Z Gold: The new Goldent Oper Dent 1996;21:36–39.
2. Balzig P, Gagro D. The use of gold foil in restorative dentistry. Acta Stomatol Croat 1986;20:239–42.
3. Clark LL. Advantages and Disadvantages of direct gold restorations. Oper Dent 1985;10–27.
4. Germain HAST Jr, Rusz JE Jr. Restoring Class VI abrasion/erosion lesions with direct gold. Oper Dent 1996;21:49.
5. Henry D. Gold foil restorations: An old friend whose time has come again. Todays FDA 2001;13:24–28.
6. Leach CD, Grubb R. Diamond working faces on gold foil condenser points. Oper Dent 1983;8:42–45.
7. Margetis M. Gold foil restorations. JADA 2002;133:686.
8. Medina J. Mat gold-Gold foil restoration. Oper Dent 2008; 33:239–45.
9. O'Connel MA. Gold foil: Is it a health hazard or a superior material? Dent Stud 1980;59:41.
10. Richter WA, Mahler DB. Physical properties vs. Clinical performance of pure gold restorations. J Prosth Dent 1973; 29:434–37.
11. Stibbs GD. Direct gold in dental restorative therapy. Oper Dent 1980;5:107–10.